

Predicting Car Prices Using Vehicle Characteristics

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Introduction

In this observational study, we will be exploring how different vehicle characteristics influence car prices using a dataset found on the Kaggle Website. The dataset was produced by Cooperunion and Jeffrey Shih in 2017, based on information released by Edmunds, who seems to be car tweeterian. We selected this dataset of cars because we wanted to explore what drives the change in car prices in the car market and society, particularly, whether cars' characteristics are significant factors in determining the variation of car prices, and whether the year of production affects car prices. Accessing car prices is relevant because it is the premise of online car trading, and a reasonable price can reflect the objective, fair, and true nature of the car market. The big question for our analysis: can we predict cars' prices based on cars' characteristics? To answer this question, I will split our model into two different models, with a single indicator, as we interpret the outputs in the results section. Specifically, we will be answering two different questions, such as, Is the relationship between price and engine the same for cars produced before 2001 as for those produced after 2001? Is the association between car prices and the average fuel efficiency of cars the same for vehicles produced before as for those produced after 2001?

To place our work in the context of existing literature, we reviewed two relevant studies. Liu et al. (2022) analyzed used-car prices using a combination of neural networks and optimization methods and found that engine characteristics, transmission type, and mileage were consistently strong predictors of price. Their findings support our expectation that mechanical features play a key role in determining vehicle value. A second study by Bukvić et al. (2022) applied supervised machine-learning methods to predict used-vehicle prices and found that linear regression performed extremely well, achieving an R^2 of 0.95 when using variables such as mileage, year, and engine specifications. This reinforces the suitability of regression-based approaches for modeling car prices using feature-level data.

Based on the dataset and these prior studies, we formed several hypotheses. First, we hypothesize that highway MPG and city MPG will be negatively associated with MSRP, since more fuel-efficient vehicles tend to be smaller or less performance-oriented. Second, we expect that engine-related characteristics will be positively associated with price, reflecting the higher costs and performance capabilities of larger or more powerful engines. Third, we hypothesize that automatic transmission vehicles will have higher prices than manual vehicles, and that vehicle size will be positively associated with MSRP. Finally, we anticipate that the relationship between engine characteristics and price may differ between vehicles produced before and after 2001.

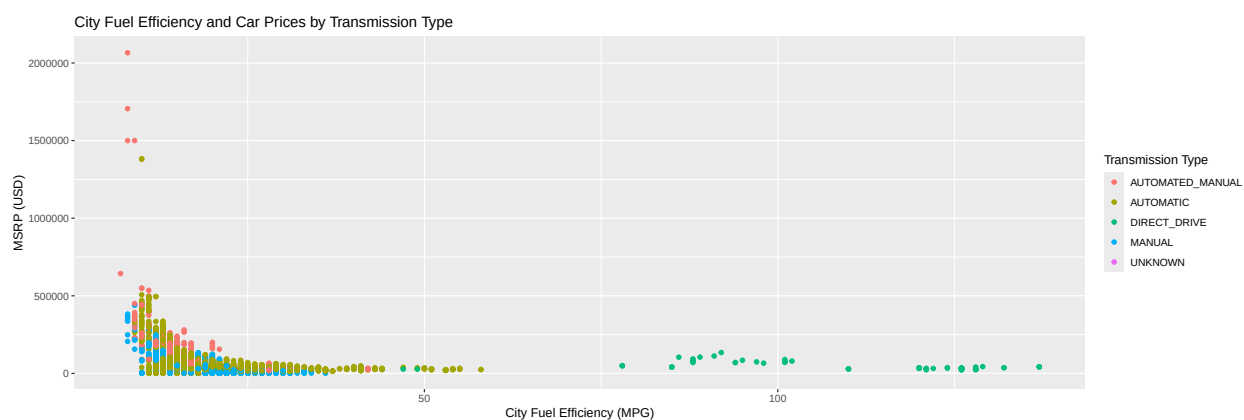
In the sections that follow, we describe our materials and methods, including data cleaning decisions and modeling strategies, present the results of our exploratory data analysis and regression models, and conclude with a discussion of our findings, limitations, and directions for future research.

Materials and Methods

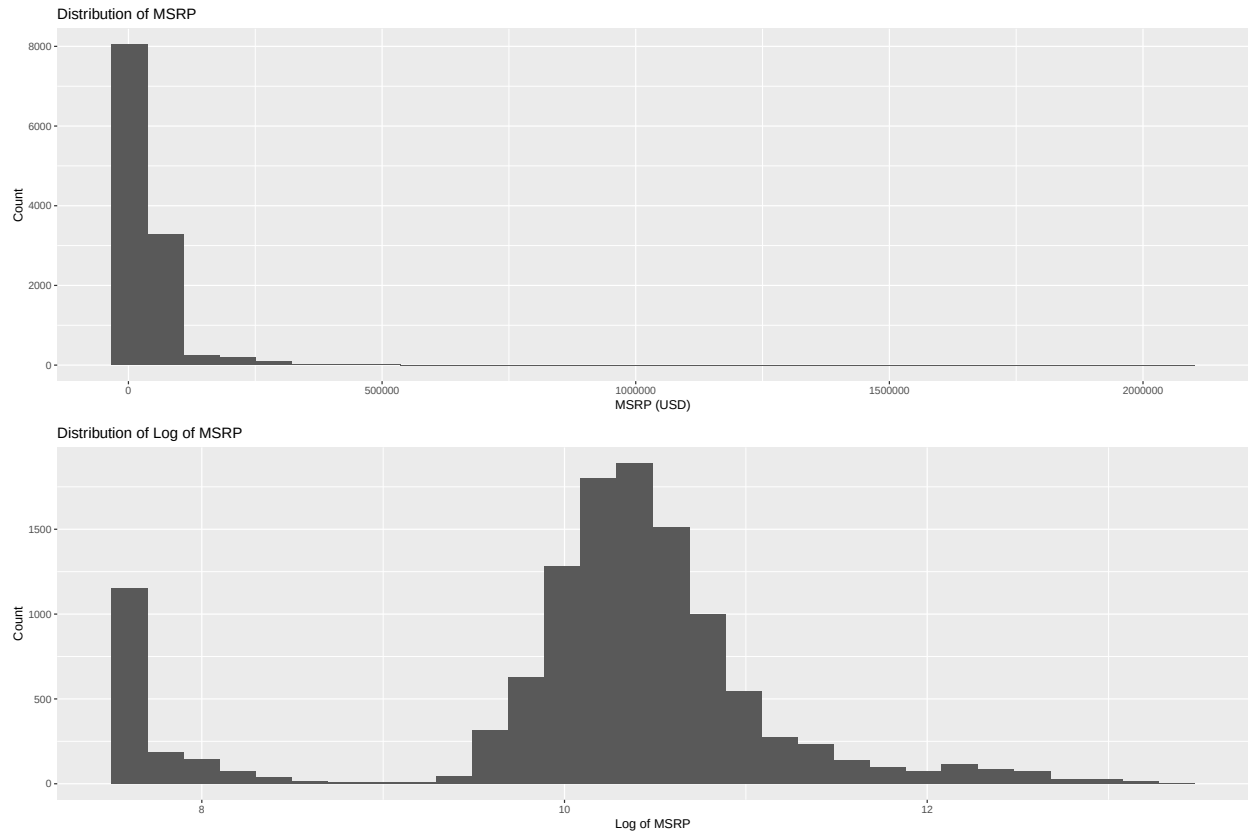
The dataset contained 11,914 observations and was produced by Cooperunion and Jeffrey Shih in 2017, based on information released by Edmunds, who seems to be car tweeterian. The dataset is divided into car maker, car model, release year, engine fuel type, engine horsepower, number of engine cylinders, transmission

type, wheel position, number of doors, market category, vehicle size, vehicle style, highway mpg, city mpg, popularity and manufacturer's suggested retail price. Each row corresponds to each car. We decided to use retail price (MSRP) in place of the car price. Based on internet search, the msrp can be represented as \$. Release year is between 1990 and 2017. There are 5 transmission types stated below. There are 10 engine fuel types, the largest proportion being regular unleaded by 60%. However there are too many variable types, separated by wording such as premium or recommended. Our understanding in the engine system is limited to properly cleaning and reorganizing the category. More than half of cars are automatic (69%). Vehicle size is categorical: compact, large, midsize. In order to predict MSRP, we will use mpg variables and engine horsepower. However, there is concern about engine horsepower resembling more of a character variable rather than a numerical variable.

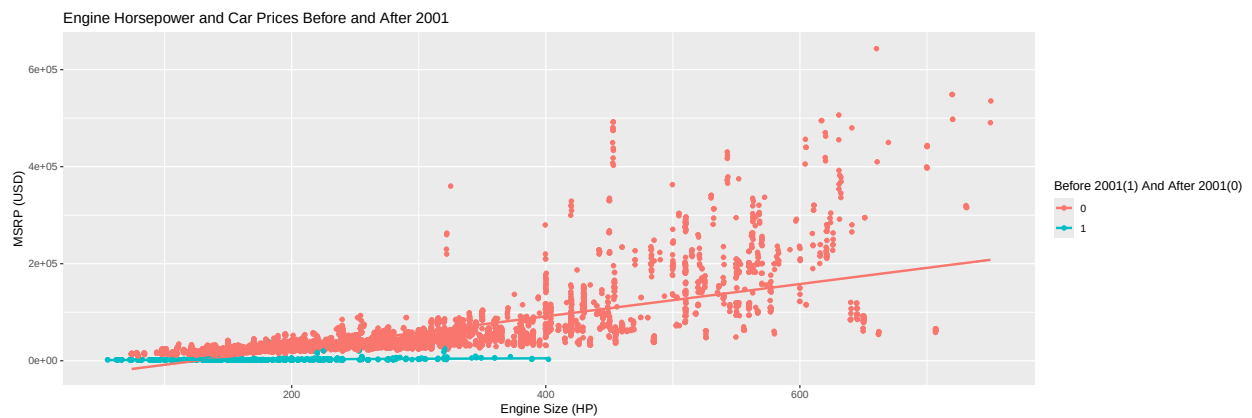
Due to the right-skewed distribution of data, we embarked on a data cleaning process. Under MSRP statistics analysis on the distance between max and the mean (2065902 to 40594.74), I removed cars with msrp above \$1000000. One of the removed cars, the Lamborghini Reventon, is a supercar with only 40 releases. It seems such cars would be considered luxury and not representative of the relationship between car features and price in general. Based on the graph relating two mpgs on msrp, we excluded Audi A6 (2017) as it had highway mpg of 314, two times above the next highest car. Meanwhile, the graph showed cars with electric engines have much better mpg performance compared to other cars. Because electric cars would have been relatively new in 2017, and that its car features seem to be analyzed more like IT products compared to traditional cars, it seems reasonable to remove electric cars from the dataset. Also I removed cars with unknown transmission types.

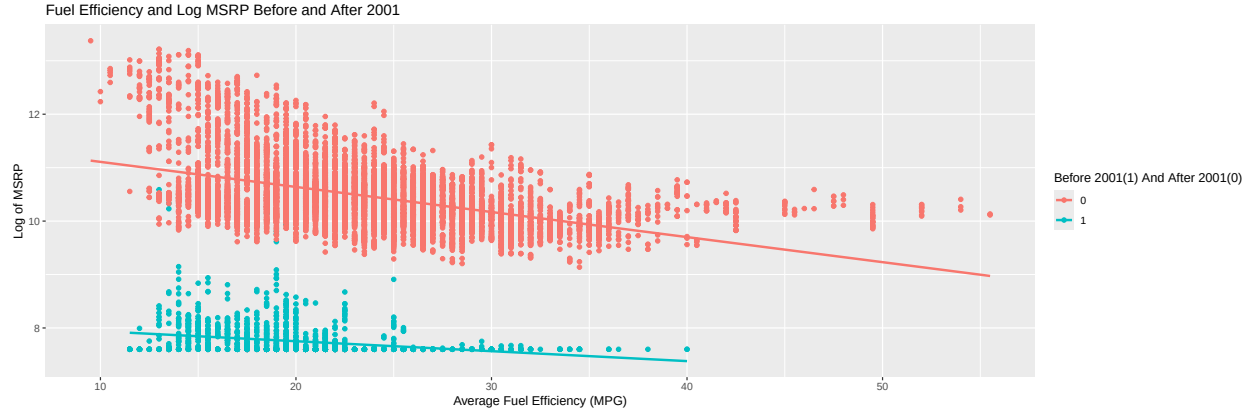


Based on strong correlation between highway and city mpg (0.8868), and for speedy analysis, the mpg will be measured as a mean of two values. Meanwhile, we chose to log MSRP since we are concerned about the right skewed distribution of data. Therefore, our interpretation of the result would be on increase of median MSRP by factor of eb per each variable value increase. This decision may limit the ability to understand the influence of engine horsepower and mean mpg on the MSRP. On the other hand, this way of interpretation could be beneficial in understanding the differences between pre and past 2001 cars.

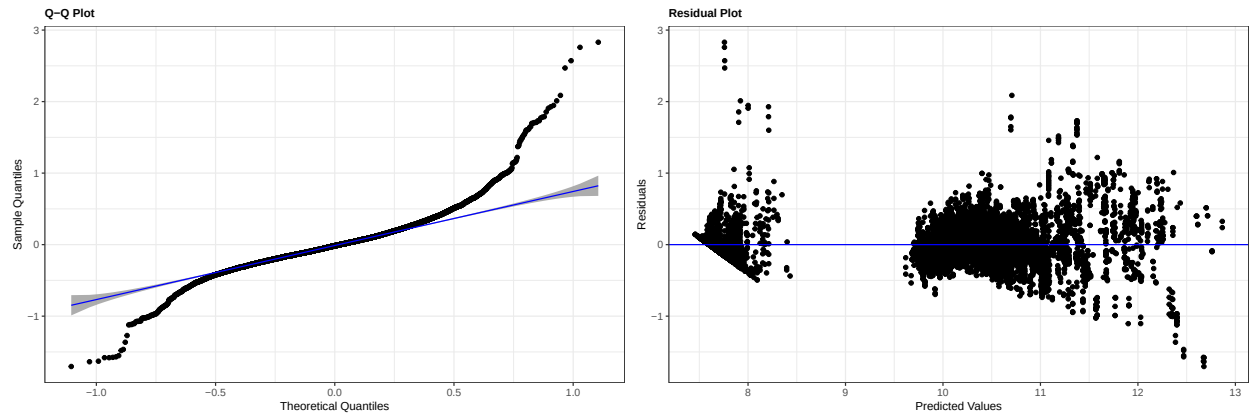


As mentioned above, we are interested in the characteristic differences between cars released before and after 2001. The plots for mean mpg vs $\log(\text{MSRP})$ and horsepower vs $\log(\text{MSRP})$ (located below) show a clear distinction between cars before and after 2001. Therefore, we believe multiple regression models with interaction terms is appropriate.





We will create 3 linear regression models. The first model will predict $\log(\text{MSRP})$ based on mean mpg, categorical variable for cars before and after 2001, and interaction term. The second model will predict $\log(\text{MSRP})$ based on engine HP, categorical variable for cars before and after 2001, and interaction term. The last model will predict $\log(\text{MSRP})$ with both numerical variables, categorical variable, and interaction terms for each numerical variables. We produced residual panels for each models and although we are somewhat concerned with the shape of the q-q plot, we believe the regression conditions are satisfied. We speculate that distinctive characteristics of pre-2001 cars would have caused curvature at the end of the q-q plot. Although it seems possible that a dataset with post-2001 cars only would have better linearity, we decided to keep past-2001 cars as they are part of our research, and because they represent roughly 10% of the entire dataset. Below is the qq plot and residual plot for the final model.



Results

In our model, we examined the relationship between price (MSRP) and the mean fuel efficiency (MPG), and engine size (HP), where our indicators are cars produced before 2001 and those produced in or after 2001. The overall model for analysis would be: $\log(\hat{MSRP}) = \beta_0 + \beta_1(\text{mean}_{mpg}) + \beta_2(\text{enginesize}) + \beta_3(I2001) + \beta_4(\text{mean}_{mpg} : I2001) + \beta_5(\text{enginesize} : I2001) = 7.1343 + 0.0048(\text{mean}_{mpg}) + 0.0031(\text{enginesize}) + 1.819(I2001) + 0.0045(\text{mean}_{mpg} : I2001) + 0.00195(\text{enginesize} : I2001)$. In this model, we wanted to see whether the relationship between price and the average fuel efficiency is the same for cars produced before 2001 as for those produced in or after 2001. We examined the relationship between price and engine size in the same way for cars produced before 2001 and in or after 2001. We divided cars into two groups: those produced before 2001 and those produced after 2001, as cars produced before 2001 are considered 20+ old, we therefore are curious about whether there is a predicted price difference. We incorporated interaction terms into our multiple linear regression model to check if these associations are statistically significant. To easily understand the effects of each coefficient on the response variable, we interpreted our model as two

different models based on the two main predictors, the mean fuel efficiency (MPG) and engine size (HP), and the indicators.

The model shows a difference and positive relationship between price (dollar) and the average fuel efficiency (MPG) on both types of cars, those produced before and after 2001. The fitted model equation for this association: $\log(\hat{MSRP}) = \beta_0 + \beta_1(\text{mean}_{mpg}) + \beta_2(I2001) + \beta_3(\text{mean}_{mpg} : I2001) = 7.1343 + 0.0048(\text{mean}_{mpg}) + 1.819(I2001) + 0.0045(\text{mean}_{mpg} : I2001)$. For cars produced before 2001 (estimate = 0.0048, p-value = 0.0126), for every 1-MPG increase in the average fuel efficiency, the median of price is multiplied by a factor $e^{0.0048} = 1.0048$, holding other predictors constant. In contrast, for cars produced after 2001 (estimate = $0.0048174 + 0.0045187 = 0.0093$), for every 1-MPG increase in the average fuel efficiency and holding other predictors constant, the median of car price is multiplied by a factor $e^{0.0093} = 1.0093$. This difference in multiplicative factor in predicting car price based on the average fuel efficiency for both types of car ($1.0093 - 1.0048 = 0.0045$), indicated by a significant of (p-value = 0.0253) of the interaction term, is statistically significant.

The relationship between car price (dollar) and engine size (HP) also appear positive and different for cars produced before and after 2001. The fitted model equation for this association: $\log(\hat{MSRP}) = \beta_0 + \beta_1(\text{enginesize}) + \beta_2(I2001) + \beta_3(\text{Enginesize} : I2001) = 7.1343 + 0.0031(\text{Enginesize}) + 1.819(I2001) + 0.00195(\text{Enginesize} : I2001)$. The model reveals a positive correlation between price and engine size for cars produced before 2001 (estimate = 0.0030, p-value = $<2e-16$). For every 1-HP increase in the engine size, the median of car prices is multiplied by a factor $e^{0.00308} = 1.0031$, holding other predictors constant. In contrast, for vehicles produced after 2001 (estimate = $0.00308 + 0.00195 = 0.00503$), each time there is a 1-HP increase in the engine size, the median of car price is multiplied by a factor $e^{0.00503} = 1.0050$, holding other predictors constant. This different multiplicative factor in predicting car price based on engine size for both types of car ($1.0050 - 1.0031 = 0.0019$) is statistically significant as according to the p-value of the interaction term (p-value = $<2e-16$).

The full multiple linear regression model has an R-squared value of 0.9349 and an adjusted R-squared of 0.9348, suggesting a strong explanatory power. All of these variables together explain 93.49% of the variability in car prices for different cars. The residual standard error is 0.2813 on 11788 degrees of freedom indicating small unexplained variation by the model in predicting prices. Overall, this model is statistically significant (F-statistic: $3.384e+04$ on 5 and 11788 DF, p-value: $< 2.2e-16$).

Discussion

Our multiple linear regression model examined whether the relationships between car price and two key vehicle characteristics, fuel efficiency (MPG) and engine size (HP), differ for cars produced before 2001 compared with those produced in or after 2001. The results indicate that both fuel efficiency and engine size are positively associated with car prices, but vehicles produced after 2001 show notably stronger associations. According to the outputs and all the significant p-values of the parameter estimates, the mean fuel efficiency and engine size are considered significant predictors of car prices, yet car's version also has significant impact on car's prices.

What we found interesting about our analysis was seeing how much the model changed from a model without logging the price variable to a model with logging the price variable. Fitting the same model without logging the price variable, we got a relatively small R-squared value of 0.57, engine size and mean fuel efficiency are statistically significant predictors of car prices as their p-values are very big. In contrast, after logging the price variable, our R-squared value became 0.93 and all the explanatory variables became significant with small p-values. Regarding the interpretation, it is interesting to see that car characteristics impact prices of older and newer cars differently, which is consistent with our initial assumption.

The stronger association between fuel efficiency and price among newer cars likely reflects changes in consumer preferences, regulatory standards, and technological advancements over the past two decades. Rising fuel prices, greater environmental awareness, and stricter emissions regulations may have increased the value consumers place on fuel efficiency, especially in newer vehicles where efficiency gains are often paired with

advanced technology and improved performance. Similarly, the larger price premium associated with engine size in newer cars may reflect improvements in engine design that allow higher horsepower without sacrificing reliability or efficiency, making additional power more attractive and valuable to consumers.

Factors such as vehicle brand, market category (for example, luxury versus economy), number of doors, transmission type, and technological options were not explicitly controlled for in this analysis but are likely correlated with both vehicle price and the predictors of interest. Additionally, newer cars may systematically differ from older cars in unmeasured ways, such as maintenance history or market availability, which could partially explain the stronger associations observed for after-2001 vehicles.

In terms of generalizability, we could apply these findings to a similar situation like predicting the median prices for a general combination of cars, but not a specific type or brand of cars, because the dataset includes cars from 46 different manufacturers, encompassing a wide range of vehicle classes, price levels, and design characteristics. As a result, the estimated relationships between price, fuel efficiency, and engine size reflect an average effect across many types of vehicles rather than effects specific to particular market segments. Different categories of vehicles, such as luxury cars, economy vehicles, trucks, or performance, oriented models, may value fuel efficiency and engine size differently, meaning that the observed associations may not generalize equally across all vehicle types.

A few things that we considered as limitations in our analysis, first, we used only one single indicator to represent vehicle age may oversimplify complex changes in automobile design and market dynamics over time. Second, two explanatory variables may not be sufficient in determining the true change in car prices. Third, the limitation of accessing the details of the dataset, particularly how the data was collected, whether there is bias, and the specification of variables, like they were measured, for instance, fuel highway (MPG). The implications of these limitations could affect the complex changes in automobile design and consumer preferences over time, with the use of just a single indicator. Limiting the model to only engine size and fuel efficiency reduces its ability to fully explain variation in car prices, as important factors such as brand, vehicle condition, and technological features are not included. Additionally, limited exploration of alternative model specifications and incomplete information about data collection and variable measurement may introduce uncertainty and potential bias in the estimated relationships.

Despite these limitations, the analysis shows good correlations, the sample size is large enough and allows for precise estimation of relationships. The use of interaction terms enables a clear comparison of how associations differ across vehicle production periods, and the log-transformation of price facilitates interpretable, proportional effect estimates. Together, these features provide a robust framework for understanding how the valuation of vehicle characteristics has evolved over time. Future research could build on this work by incorporating additional vehicle attributes, such as brand, transmission type, vehicle size driven wheels, and number of doors, to better isolate the effects of fuel efficiency and engine size. Examining narrower production periods or using a continuous measure of vehicle age may also provide a more nuanced understanding of how market valuations change over time. Finally, longitudinal data tracking prices of similar models across years could help disentangle cohort effects from broader market trends and strengthen causal interpretation.

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